Barriers to International Student Mobility

Survey Analysis – Four-Group Approach

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General Information

Sample

- **1230** total respondents
- **808** usable ($\geq$ 70% of key variables)
- Median completion rate: **68%**

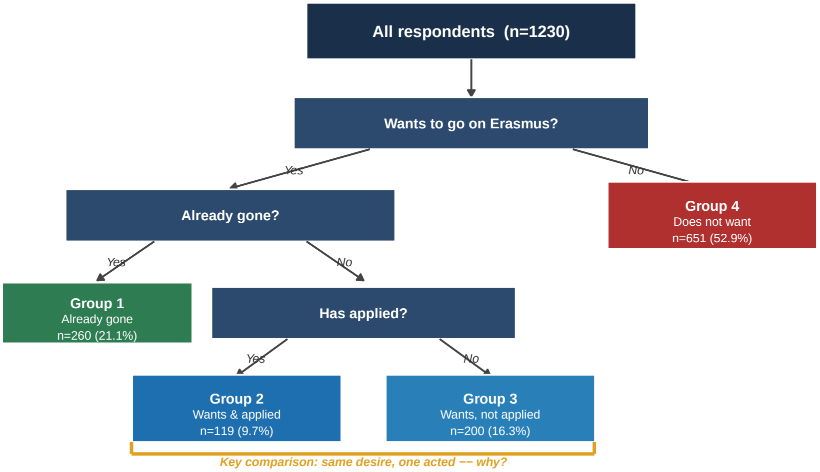
Four Groups

1. Already gone on Erasmus
2. Wants to go & has applied
3. Wants to go, has not applied
4. Does not want to go

Variables

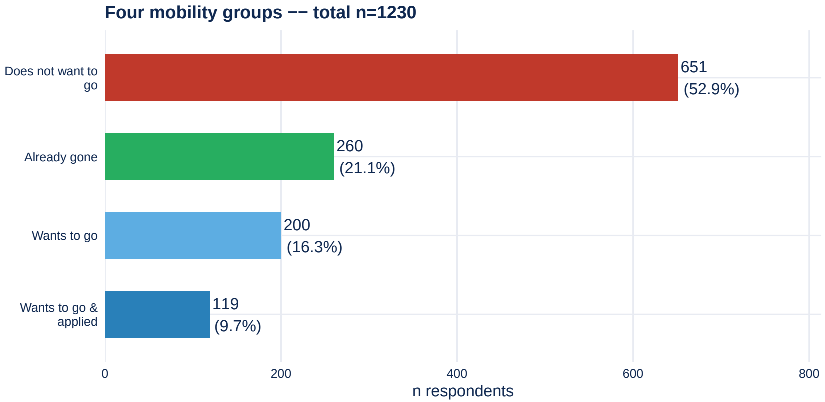
- 16 barriers (Likert 1–5)
- 11 reasons for going
- 27 psychological profile items
- Socio-demographic variables
- **New:** Parental Erasmus (Q13), Certified English (Q24)

How Are the Four Groups Defined?



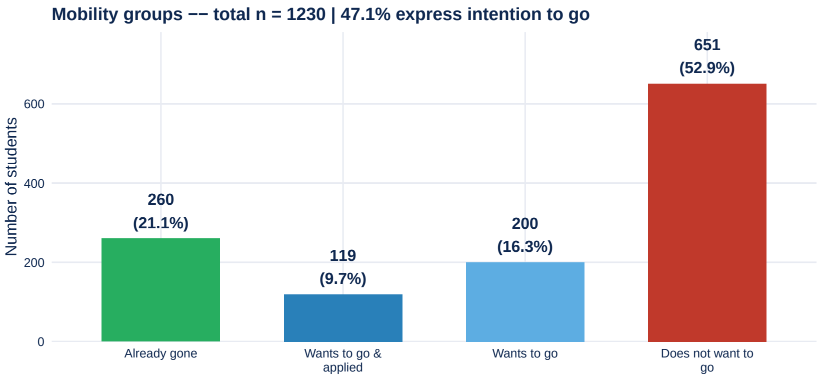
Decision tree: each student is assigned to exactly one group based on three binary questions. Groups 2 and 3 share the same desire for mobility – the key analytical question is what differentiates them.

The Four Mobility Groups



Reading: each bar = number of students. Focus on Groups 2–3 (both want to go): the gap reveals how many are blocked at the application stage.

Departure Rates and Conversion Funnel



47.1% of respondents express some intention to go abroad. Among those who want to go, the conversion to application is the key lever studied.

Analytical Framework

Two Central Questions

- **Who** are the four groups? (socio-demo, academic, financial, psychological)
- **What** differentiates students who applied from those who want to go but haven't?

Three Mechanisms

1. **Structural**: finances, grades, parental capital
2. **Psychological**: international profile, openness
3. **Informational**: lack of knowledge about procedures

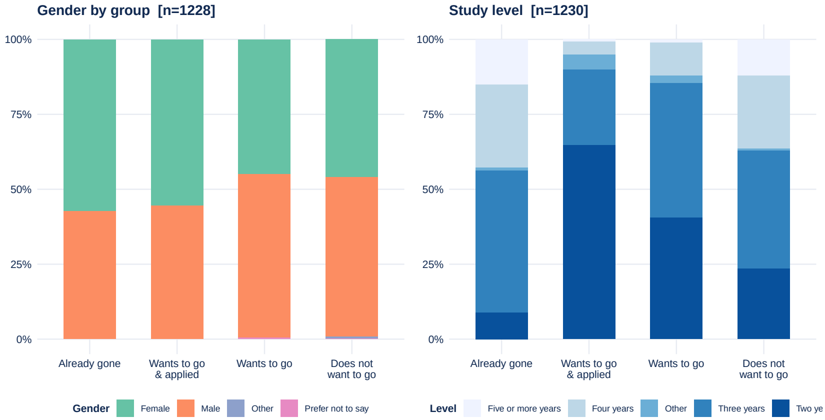
Key Comparison

Groups 2 vs 3: same desire, but only one has taken action.
What makes the difference?

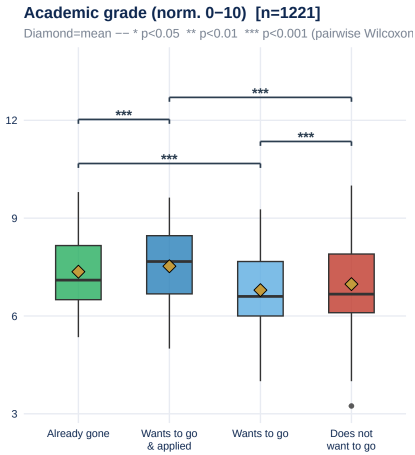
Methods

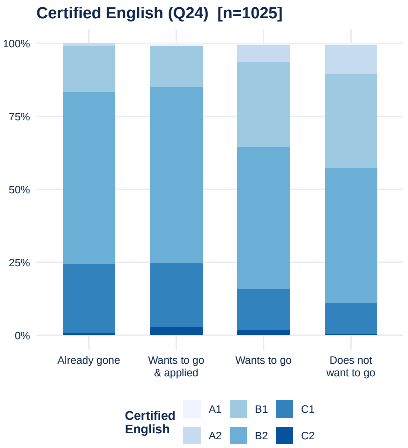
- Descriptive by group
- MCA + composite barrier score
- OLS on score
- Logistic regressions

Socio-Demographic Profile by Group



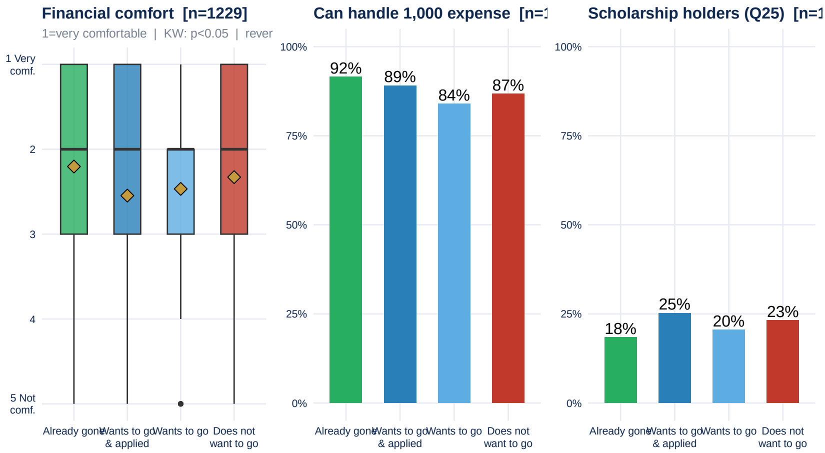
Reading: stacked bars show group composition. Look for visible differences in gender or study-level distribution – these signal structural selection into mobility.





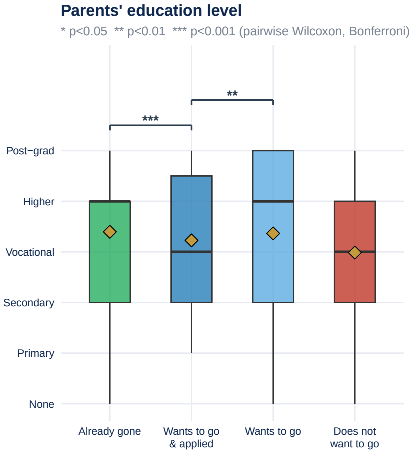
Reading: box = middle 50% of values, diamond = mean. A higher mean grade in Group 1 (already gone) would suggest academic selection into Erasmus participation.

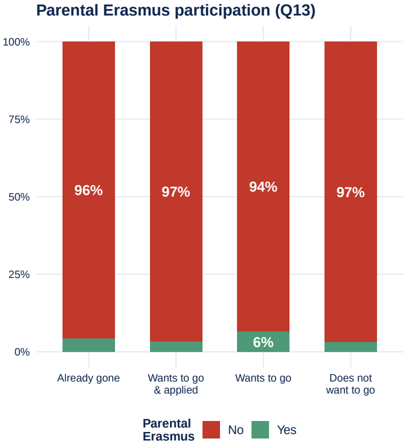
Financial Profile by Group



*Left: financial comfort scale reversed (1=very comfortable). Kruskal-Wallis global test reported in subtitle. Significant pairs (Bonferroni): Already gone vs Wants to go *. Middle: share able to absorb a 1,000 unexpected expense. Right panel: share of scholarship holders (Q25) by group – higher scholarship rates in constrained groups would confirm structural financial barriers.*

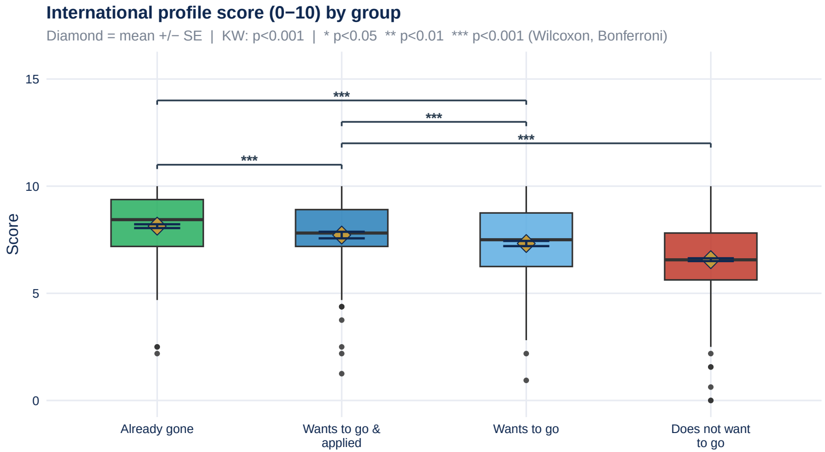
Parental Capital by Group





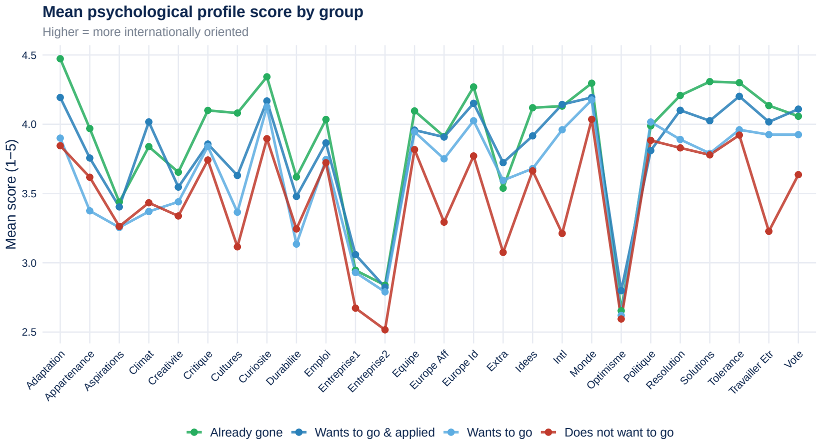
Reading: left = parents' education level (higher = more cultural capital). Right = parental Erasmus experience provides direct knowledge of the programme and process.

International Profile by Group



Kruskal-Wallis across 4 groups: $p<0.001$. Pairwise brackets show Bonferroni-corrected Wilcoxon tests (significant pairs only). Higher score = more open to international mobility.

Psychological Profile: Group Differences



Reading: each line traces mean scores across 27 psychological dimensions. Groups 1–2 scoring higher on openness/international dimensions than Group 4 would confirm the psychological mechanism.

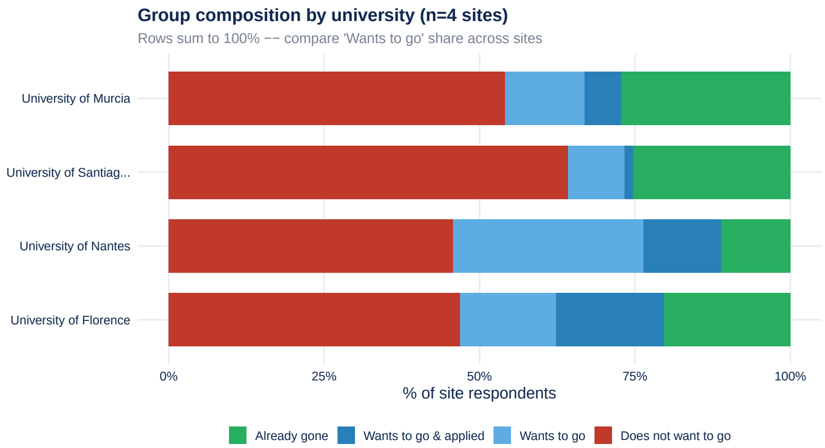
Are Group Differences Statistically Significant?

Table 1: Group differences: statistical tests

Variable	Test	Statistic	p-value	Sig.
genre	Chi-squared	17.79	0.045	Yes
niveau	Chi-squared	207.18	0.000	Yes
zone_geo	Chi-squared	43.26	0.012	Yes
parental_erasmus	Chi-squared	4.96	0.184	No
moyenne_acad_norm	Kruskal-Wallis	54.00	0.000	Yes
aisance_fin	Kruskal-Wallis	10.12	0.018	Yes
score_intl	Kruskal-Wallis	193.90	0.000	Yes
educ_num	Kruskal-Wallis	25.93	0.000	Yes

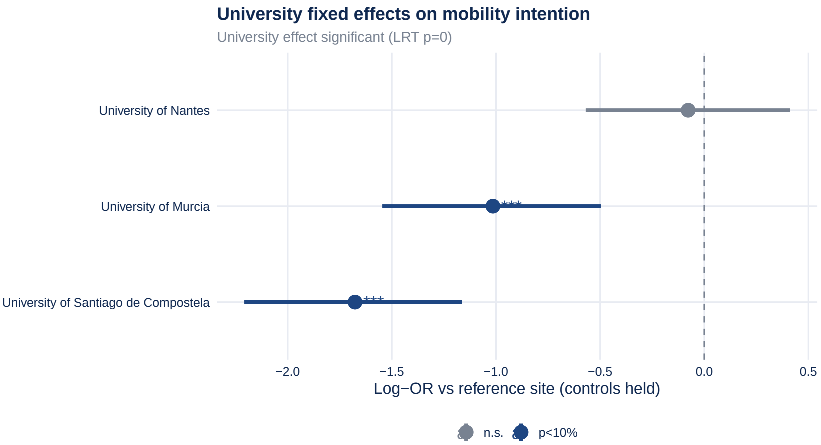
Reading: $p < 0.05$ = group difference is statistically reliable. n.s. = not significant with this sample size – not the same as no effect.

University Effect: Group Composition by Site



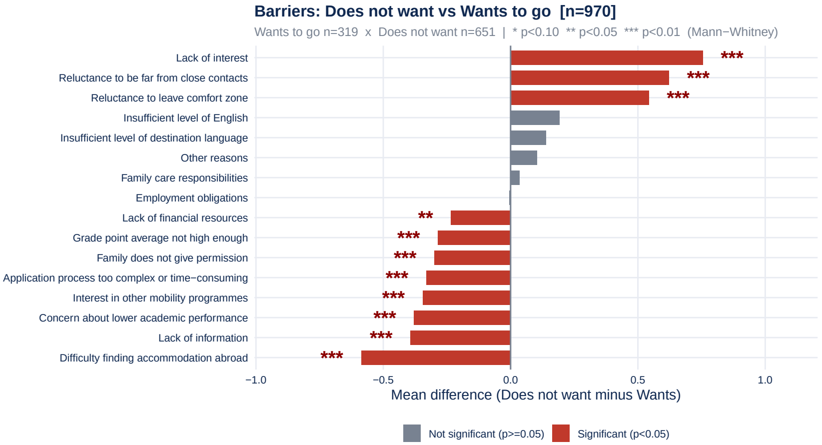
Stacked bars show the share of each mobility group per site. Sites with a larger 'Already gone' share may have a stronger Erasmus culture or more systematic information. Compare the 'Does not want to go' share to identify sites with higher structural constraint.

University Effect: Fixed Effects Logit



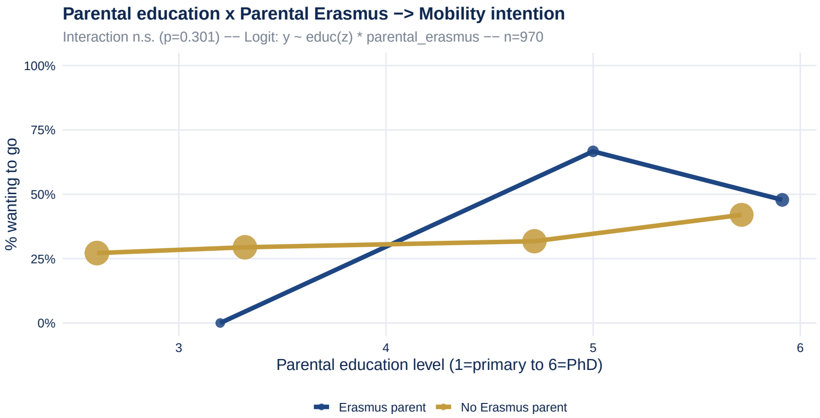
*University fixed effects after controlling for financial comfort, academic grade, parental education and international profile score. A significant LRT (p<0.05) means the site matters independently of who studies there. Coefficients are log-OR relative to the reference university (lowest alphabetical). Stars: * p<10%, ** p<5%, *** p<1%.*

Which Barriers Differentiate Wants vs Does Not Want to Go?



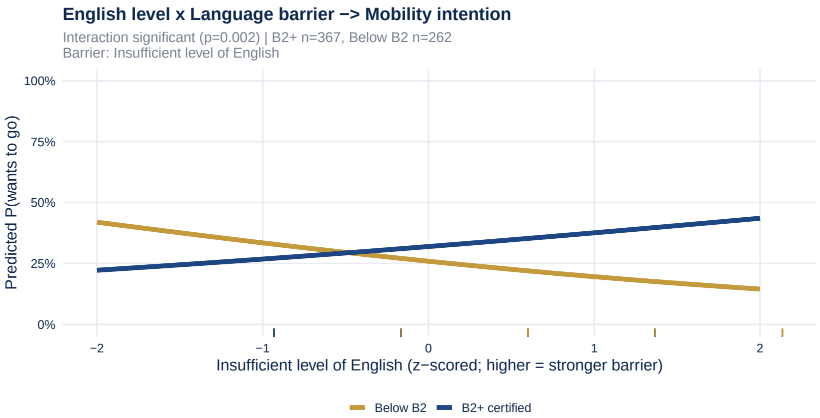
Positive bars = barriers rated higher by students who do NOT want to go. Red bars (starred) are statistically significant (Mann-Whitney p<0.05): 11 out of 16 barriers. These are the barriers that genuinely differentiate the two groups.

Interaction: Parental Erasmus x Parental Education



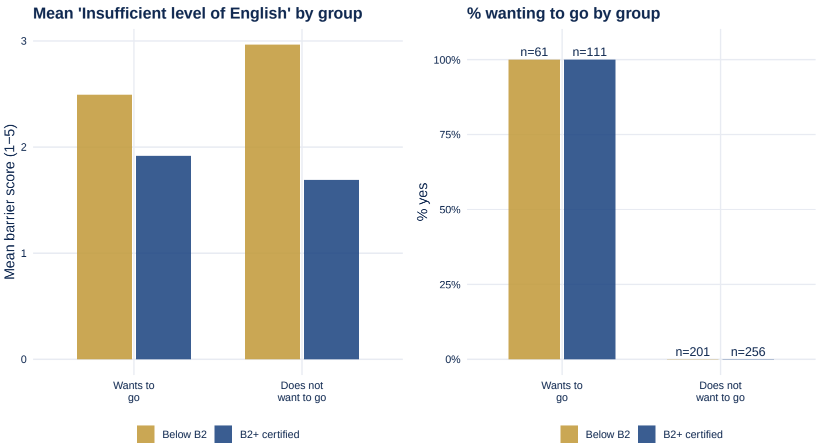
Lines show % of students wanting to go by parental education quartile, split by parental Erasmus experience. A significant interaction ($p<0.05$) would mean the education gradient is steeper (or flatter) when a parent has done Erasmus – distinguishing general cultural capital from programme-specific knowledge. Point size = cell n.

Interaction: Certified English x Language Barrier

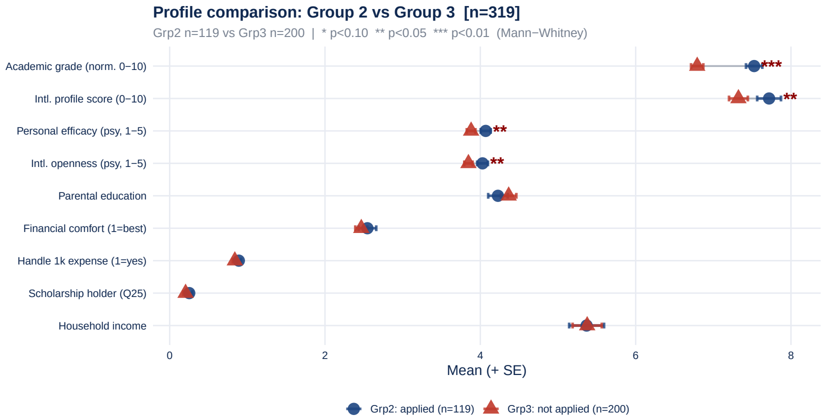


Predicted probability of wanting to go as a function of perceived language barrier, split by English certification status (B2+ vs below B2, Q24). A significant interaction confirms that certification attenuates the language barrier: certified students show a weaker negative slope between perceived language difficulty and mobility intention.

English Level x Language Barrier: Group Breakdown

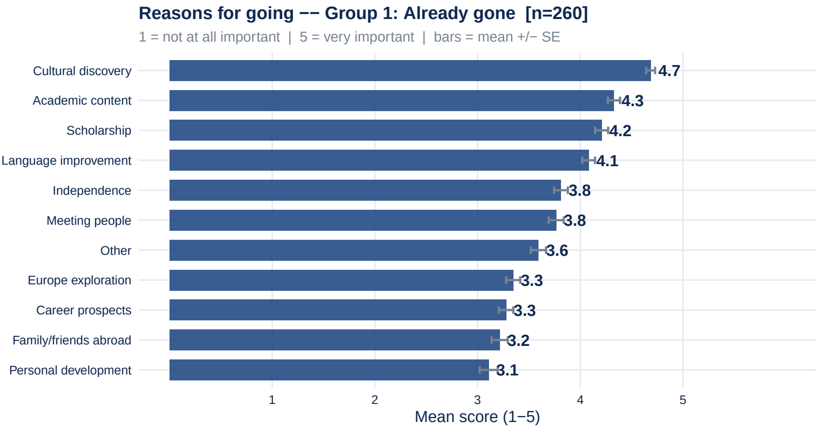


Left: mean perceived language barrier score (1=no obstacle, 5=major obstacle) by mobility group and certification level. Right: % wanting to go by group and certification level. If certified students perceive a lower barrier AND show higher mobility intention, the certification is doing double duty – both reducing perceived difficulty and boosting confidence.



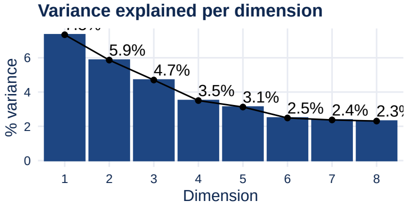
frein_ items excluded (NA for Grp2 by design). Blue circle = Grp2 (applied), red triangle = Grp3 (not applied). 4 variable(s) significantly different at p<0.05 (Mann-Whitney).

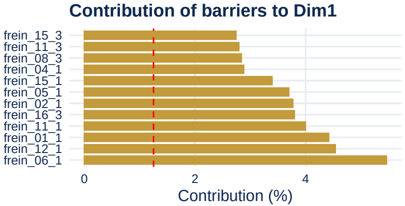
Reasons for Going – Group 1 (Already Gone)

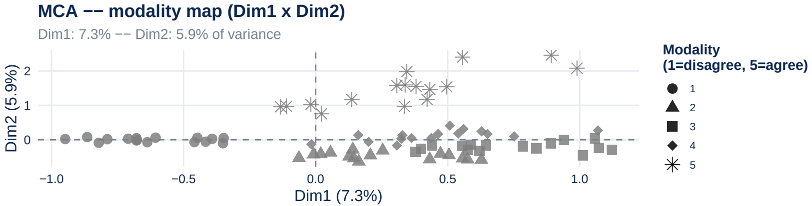


The 'reasons for going' question (raison_) is answered only by Group 1 (already gone). Other groups (2, 3, 4) did not answer this question. To compare motivations between Groups 2 and 3, refer to the barrier gap plots.

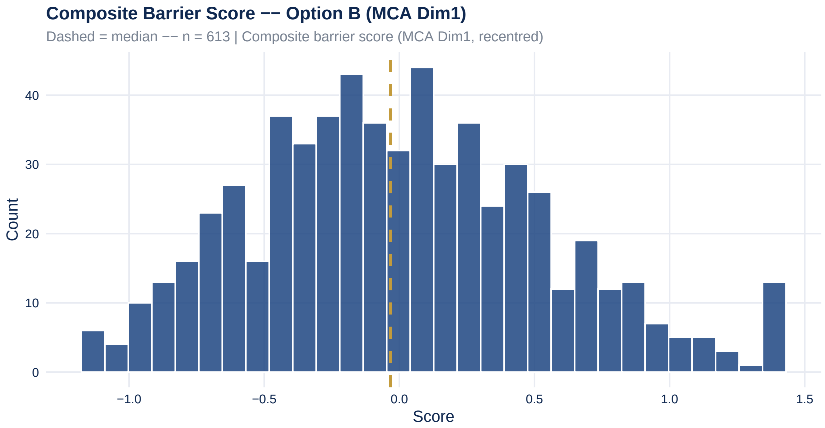
MCA on Barriers







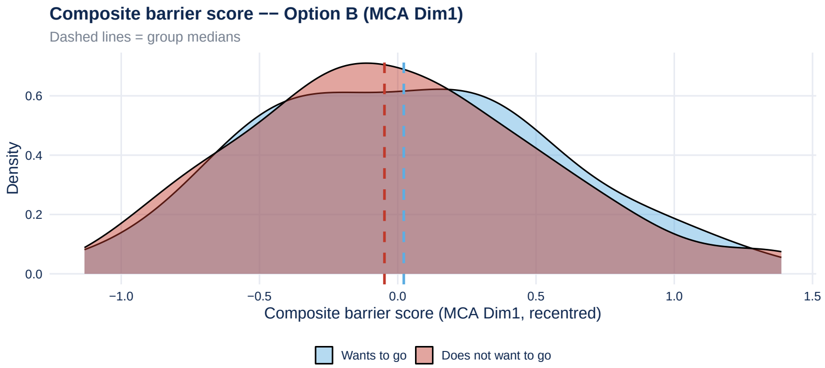
Composite Barrier Score



Scope limitation

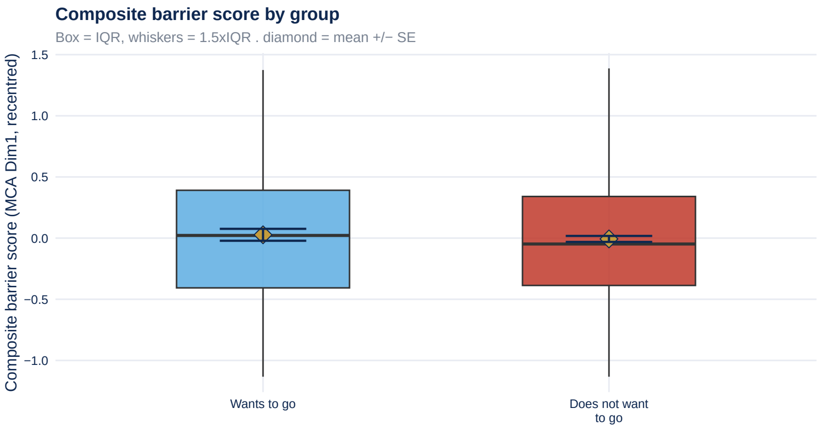
The barrier items (`frein_`) were not administered to **Group 2** (applied, $n = 119$) due to questionnaire routing: **100%** of their scores are NA. All score-based analyses (distribution, OLS, KW) therefore cover **Groups 1, 3 and 4 only**. Group 2 appears in group-level plots as an empty or near-empty curve.

Composite Barrier Score – Distribution



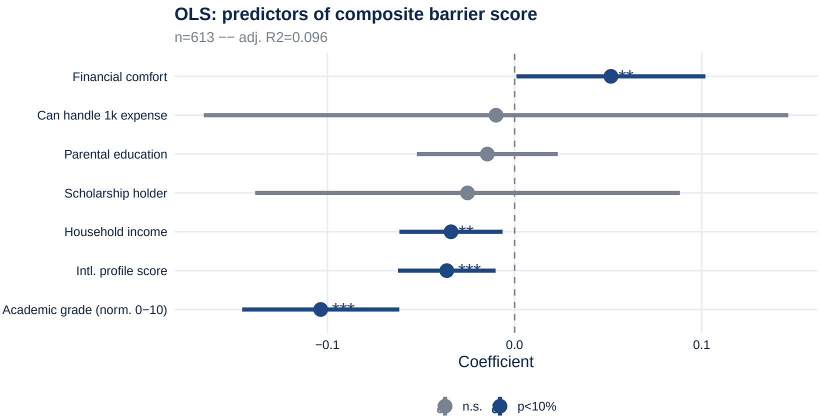
Reading: dashed lines = group medians. Curves shifted right = more barriers felt. Group 2 (applied) is absent from this plot (100% NA on barrier items). Groups 1, 3 and 4 are the analytic base for all score-based comparisons.

Score by Group: Does It Separate the Four Groups?



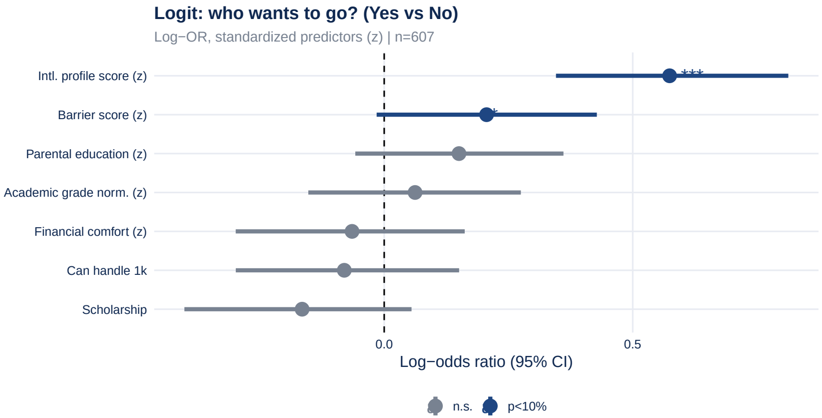
Kruskal-Wallis across 4 groups: p=0.496. Higher score = more barriers felt. ***Caution***: *Group 2 contributes only 0% non-missing scores — its box is near-empty and should not be interpreted.*

What Socio-Demo Variables Predict the Score?



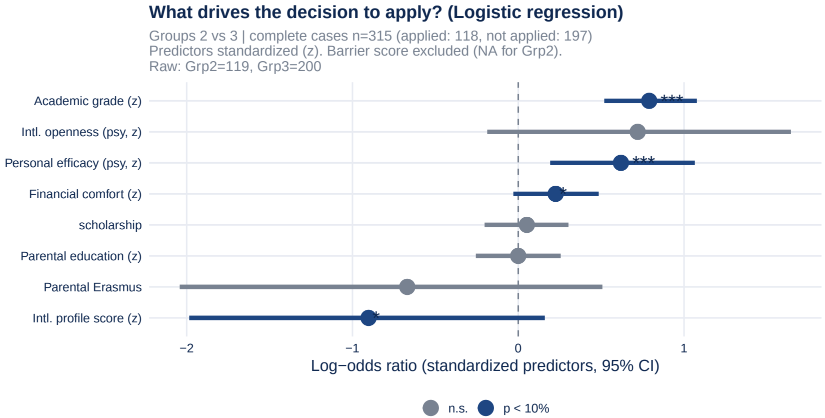
Reading: bars right = variable increases barrier score. Stars = significance (p<10%, ** p<5%). Financial comfort is coded 1 = very comfortable to 5 = not comfortable: its positive coefficient means students who are less financially comfortable perceive more barriers. Group 2 is excluded de facto (100% NA on the dependent variable).*

Logistic Regression: Movers vs Non-Movers



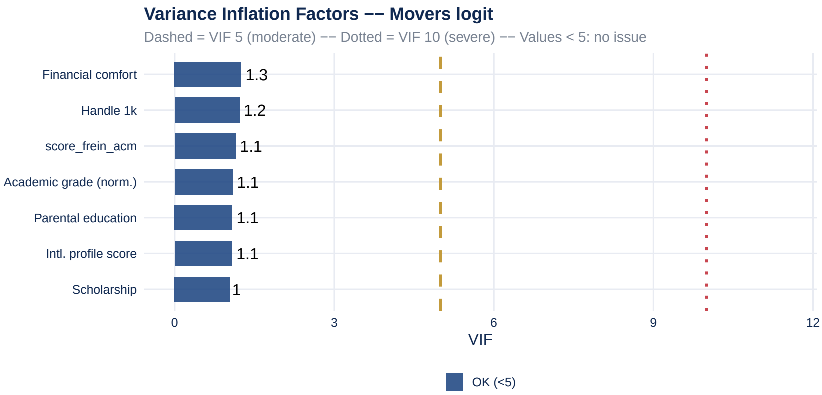
Reading: % change in odds of wanting to go. International profile is typically the dominant predictor. Significant barriers with negative effect actively deter mobility intention.

Applied vs Did Not Apply (Groups 2 vs 3)



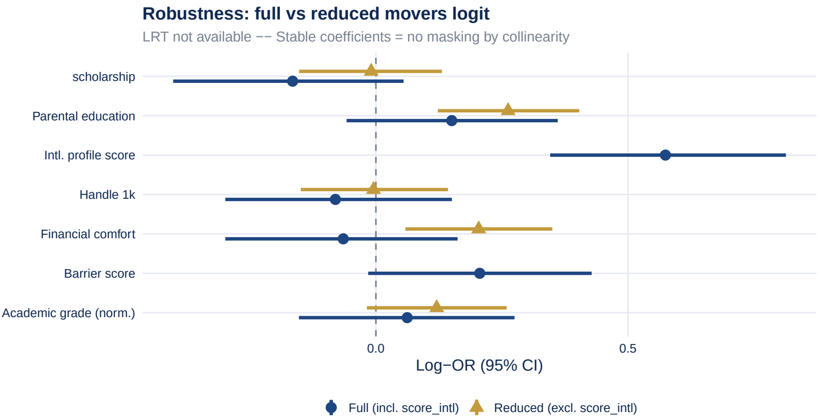
Reading: % change in odds of applying among students who want to go. Barrier score: Method A (simple mean, robust to partial NA). Same desire, different action – what tips the balance towards applying?

Multicollinearity Check (VIF)



VIF (Variance Inflation Factor) measures how much a predictor's variance is inflated by collinearity with others. VIF $> 5 =$ moderate concern; VIF $> 10 =$ severe. Here: 0 predictor(s) with VIF ≥ 5, 0 with VIF ≥ 10. Financial variables (aisance_fin, revenu_num, depense_imp) are the most likely collinear cluster; removing one or replacing all three with the structural vulnerability composite is recommended if VIF > 5.

Robustness Check: Reduced Model

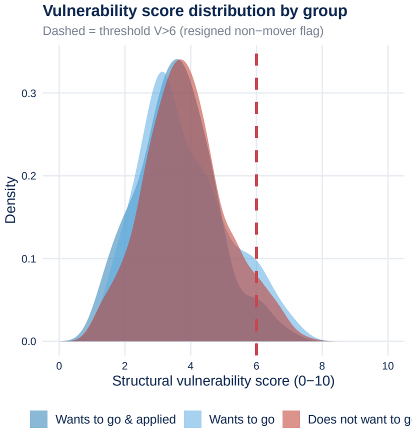


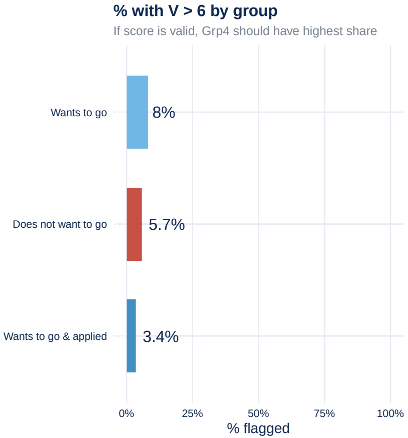
Blue = full model; gold = reduced model without `score_intl` *(international profile score). If structural coefficients (financial comfort, parental education, grade) change substantially between models,* `score_intl` *was masking them. Structural coefficients (financial, parental, academic) are stable across specifications. If the LRT is n.s.,* `score_intl` *may be dropped without loss of explanatory power for structural effects*

Distribution of the vulnerability score

Mean score by gender

Vulnerability Score: Validation





Variable	V > 6	V <= 6	n (V>6)
Financial comfort (1=best)	3.58	2.31	57
Academic grade (norm. 0-10)	6.28	7.05	56
Parental education (1-6)	2.77	4.18	57
Intl. profile score (0-10)	7.04	6.85	57
Scholarship holder (%)	49.1%	21.2%	57

Study Limitations

Methodological limits

- **Cross-sectional design**: associations only, no causal inference. Psychological profile may be shaped *by* the experience of applying, not only precede it.
- **Self-selection bias**: voluntary participation likely under-represents the most disengaged students (Grp4), biasing barrier scores downward.
- **Group 2 size**: if $n < 20$, logistic regression estimates are unstable — interpret with caution.
- **Barrier items unavailable for Grp2**: all score-based comparisons exclude applied students by design.

Measurement limits

- **Measurement invariance not tested**: psy items assumed to measure the same constructs equally across groups.
- **Likert interval assumption**: profile scores treat 1–2–3–4–5 as equally spaced; ordinal modelling not implemented.
- **Vulnerability weights (0.4/0.3/0.3)**: theoretically motivated, not empirically derived; no sensitivity analysis.
- **Grade normalisation**: assumes students report grades on their university scale — no cross-check available.

Scope

Results valid for the surveyed sample
Replication across additional cohorts

Summary of Findings

3 Mechanisms Confirmed

- **Structural**: financial comfort, parental capital, grades predict barrier score and group membership
- **Psychological**: international profile score is the strongest predictor of mobility intention
- **Informational**: lack of knowledge about procedures is independently significant

Key Finding

Grp2 vs Grp3 (applied vs not): gap driven more by **psychological profile** and **information access** than by financial constraints.

3 Action Levers

- **Info.** Clear guides + Erasmus testimonials
- **Psych.** Intercultural events from Year 1
- **Struct.** Emergency grants for constrained profiles

Limitation

$n =$ **1230** respondents. Group 2 may be small — interpret with caution. Data collection ongoing.

Recommendations by Group

Group 3 $\rightarrow$ Group 2 (lever priority)

Students who want to go but haven't applied:

- Simplify the application process
- Peer mentoring from Group 1 (already gone)
- Early information sessions (Year 1)
- Address financial uncertainty with clear grant info

Group 4 (resigned non-movers)

Students who don't want to go due to constraint:

- Build international openness through local activities
- Short mobility formats (1–2 weeks)
- Targeted financial support for vulnerability score > 6

Do not conflate

Deliberate non-movers (free choice) vs resigned non-movers (constrained). Different approaches needed.

Thank you for your attention

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Questions, comments, further analyses?

Full specifications in appendix

Appendix

Econometric Specifications

Models Estimated

1. **MCA** on barrier items (dimension reduction, 15 items)
2. **Composite score** (MCA Dim1 or simple mean)
3. **OLS** of score on socioeconomic + psychological variables
4. **Logit 1** — Yes/No mobility intention (all 4 groups)
5. **Logit 2** — Applied vs not applied (Groups 2 vs 3)

Logit specification

$$\text{logit}\ \Pr(Y_i = 1) = \alpha + \sum_j \beta_j Z_{ij}$$

Reported as odds ratios (OR). OR − 1 = % change in odds. McFadden $R^2 > 0.20$ = good fit.

Caution

$n =$ **1230**: limited power. Group 2 may have $n < 30$. Listwise deletion for all models.

Z-Score Construction

Standardisation formula

All continuous predictors in regressions are standardized:

$$Z_{ij} = \frac{X_{ij} - \bar{X}_j}{\sigma_j}$$

$\bar{X}_j$ and σ_j computed on the analytic sample.

Variables standardized (logistic models):

- Financial comfort `aisance_fin` (1–5)
- Academic grade `moyenne_acad_norm` (0–10, normalised)
- Parental education `educ_num` (1–6)
- Household income `revenu_num` (1–6)

Why standardize?

- Avoids overflow in `glm()` when scales differ
- Makes coefficients comparable: 1 unit = 1 SD shift
- Does **not** affect significance or model fit

Binary / factor predictors

`depense_imp` (0/1) and `parental_erasmus` (factor) are **not** standardized. Coefficients = log-OR for the reference change (0→1 or No→Yes).

OLS barrier score model

Predictors **not** standardized in OLS. Coefficients = effect of 1-unit change on original scale.

Profile Variables Construction

Scores & composites

Intl. profile (0–10):
`score_intl` $= \frac{\bar{x}_{8\text{ items}} - 1}{4} \times 10$
Items: `travailler_etr`, `intl`, `cultures`, `tolerance`, `europe_id`, `curiosite`, `adaptation`, `monde`

Intl. openness (1–5): row mean of `travailler_etr`, `intl`, `cultures`, `tolerance`, `europe_id`
Personal efficacy (1–5): row mean of `resolution`, `equipe`, `adaptation`, `solutions`

Grade normalisation:
`acad_norm` $= \frac{\texttt{grade}}{\texttt{max}} \times 10$
Florence /30 · Nantes /20 · others /10

Structural vulnerability (0–10)

$V_i = (0.4\,F_i + 0.3\,A_i + 0.3\,E_i)/5 \times 10$

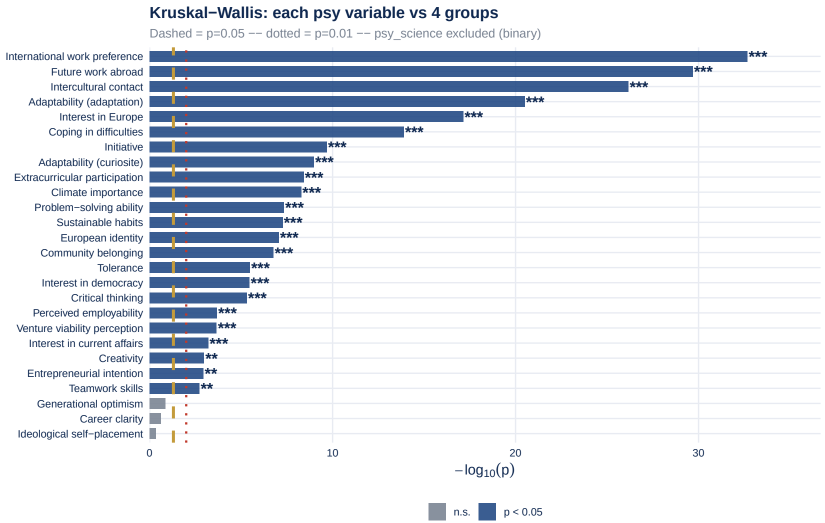
- F_i: `aisance_fin` + `revenu_num` + `depense_imp`
- $A_i = (10 - \texttt{acad\_norm})/10 \times 5$
- $E_i = (7 - \texttt{educ\_num})/6 \times 5$

Missing → 2.5. Score 0=low, 10=high constraint.
Threshold $V_i > 6$ = resigned non-mover.

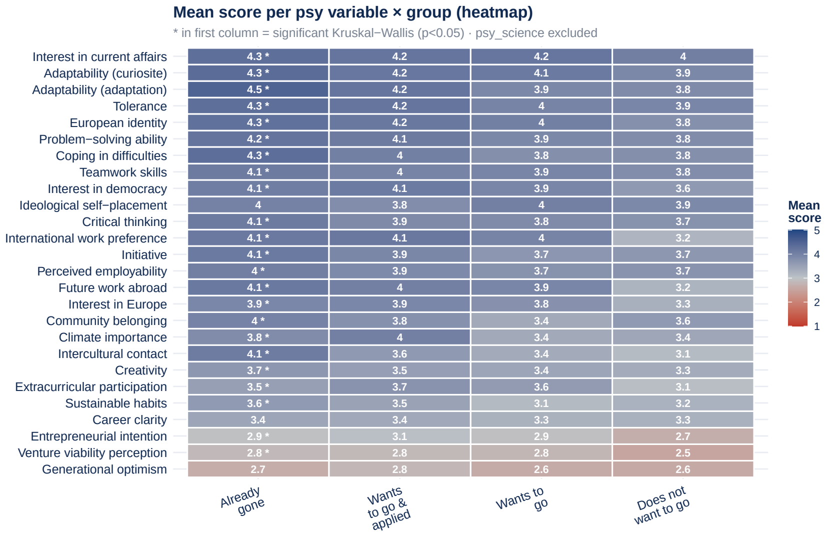
Z-score reminder

Logistic predictors standardized:
$Z = (X - \bar{X})/\sigma$.
Binary vars (`depense_imp`, `parental_erasmus`) not standardized.
OLS uses raw scales.

Groups)

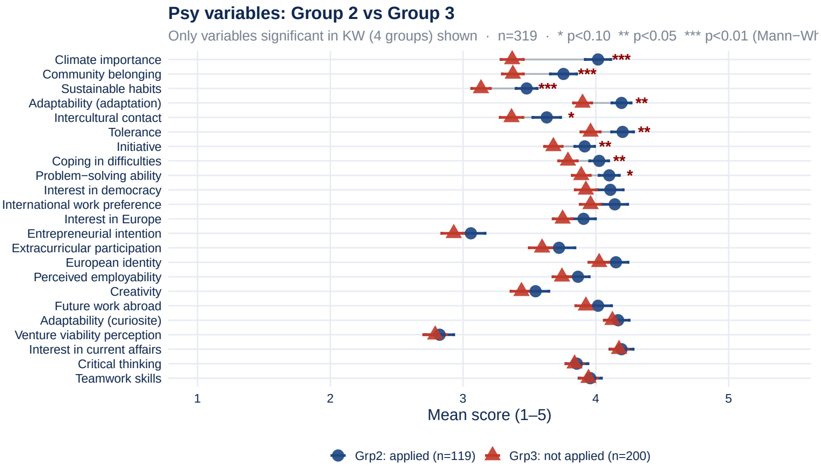


*23 out of 26 psy variables show significant group differences (p<0.05, Kruskal-Wallis). Most discriminating: International work preference, Future work abroad, Intercultural contact. Stars: *** p<0.001, ** p<0.01, * p<0.05, . p<0.10.*



*Heatmap: blue = high score (1–5), red = low score. Each cell shows the group mean. Variables marked * differ significantly across the 4 groups (Kruskal-Wallis p<0.05). Reading rows: variables with large colour contrast across groups are the most discriminating.*

Psy Variables: Groups 2 vs 3 (Significant Only)



Variables shown: 23 (significant in Kruskal-Wallis across 4 groups). Among these, 7 also differ significantly between Groups 2 and 3 (Mann-Whitney $p<0.05$). Blue circle = Grp2 (applied), red triangle = Grp3 (not applied). Sorted by absolute mean difference – most discriminating at top.